



IBM INTERNSHIP



FITNESS BUDDY

AI-Powered Personal Fitness Coach

Built with LangFlow + IBM watsonx.ai + IBM Granite

Edunet Foundation | IBM 4-Week Internship | Problem Statement #13

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Transforming Fitness with Conversational AI

STUDENT PROFILE

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Domain: Health & Fitness / AI

Project: Fitness Buddy – AI Fitness Coach



**FITNESS
BUDDY**

Problem Statement

#13

PROBLEM STATEMENT #13



THE CHALLENGE

In today's fast-paced world, many individuals struggle to maintain a healthy lifestyle due to:

- Lack of personalized fitness guidance
- Time constraints and busy schedules
- Inconsistent motivation and habit building
- Expensive gym memberships & rigid plans
- No accessible, 24/7 fitness assistant

THE OBJECTIVE

Build Fitness Buddy — an AI conversational fitness coach that:

- ✓ Recommends home workouts based on user input
- ✓ Provides motivational tips and daily inspiration
- ✓ Suggests nutritious meal ideas
- ✓ Encourages habit-building and consistency
- ✓ Remembers user preferences using Memory

TECHNOLOGY STACK



LangFlow

Visual AI workflow builder to design and orchestrate the Fitness Buddy chatbot pipeline



IBM watsonx.ai

IBM's enterprise AI platform for model deployment, scaling and AI governance



IBM Granite

ibm/granite-8b-code-instruct model via IBM watsonx.ai — mandatory IBM AI model



Message History

LangFlow Memory component for conversation history and personalized responses



Prompt Template

Dynamic prompt engineering to guide AI responses for fitness coaching behavior



IBM Cloud

IBM Cloud Lite plan infrastructure hosting watsonx.ai Runtime (Sydney region)



Approach Used: LangFlow + IBM watsonx.ai + IBM Granite Models

PROPOSED SOLUTION

Fitness Buddy is an AI-powered conversational fitness coach built using LangFlow and IBM watsonx.ai. It provides personalized home workout routines, healthy meal suggestions, daily motivation, and habit-building guidance through an intelligent chat interface — available 24/7, completely free.



Workout Plans

Personalized home workout routines based on fitness level and goals



Meal Guidance

Healthy meal ideas and nutrition suggestions for weight management



Motivation

Daily fitness inspiration and motivational tips to stay consistent



Memory

Remembers user name, goals and preferences across the conversation

LANGFLOW COMPONENTS USED

1

Chat Input

Interface where users enter fitness queries, goals, and preferences through natural language conversation

2

Prompt Template

Dynamic prompt engineering that instructs IBM Granite to behave as a fitness coach with specific rules and format

3

IBM watsonx.ai

IBM's AI model component using ibm/granite-8b-code-instruct to generate personalized fitness plans and advice

4

Message History

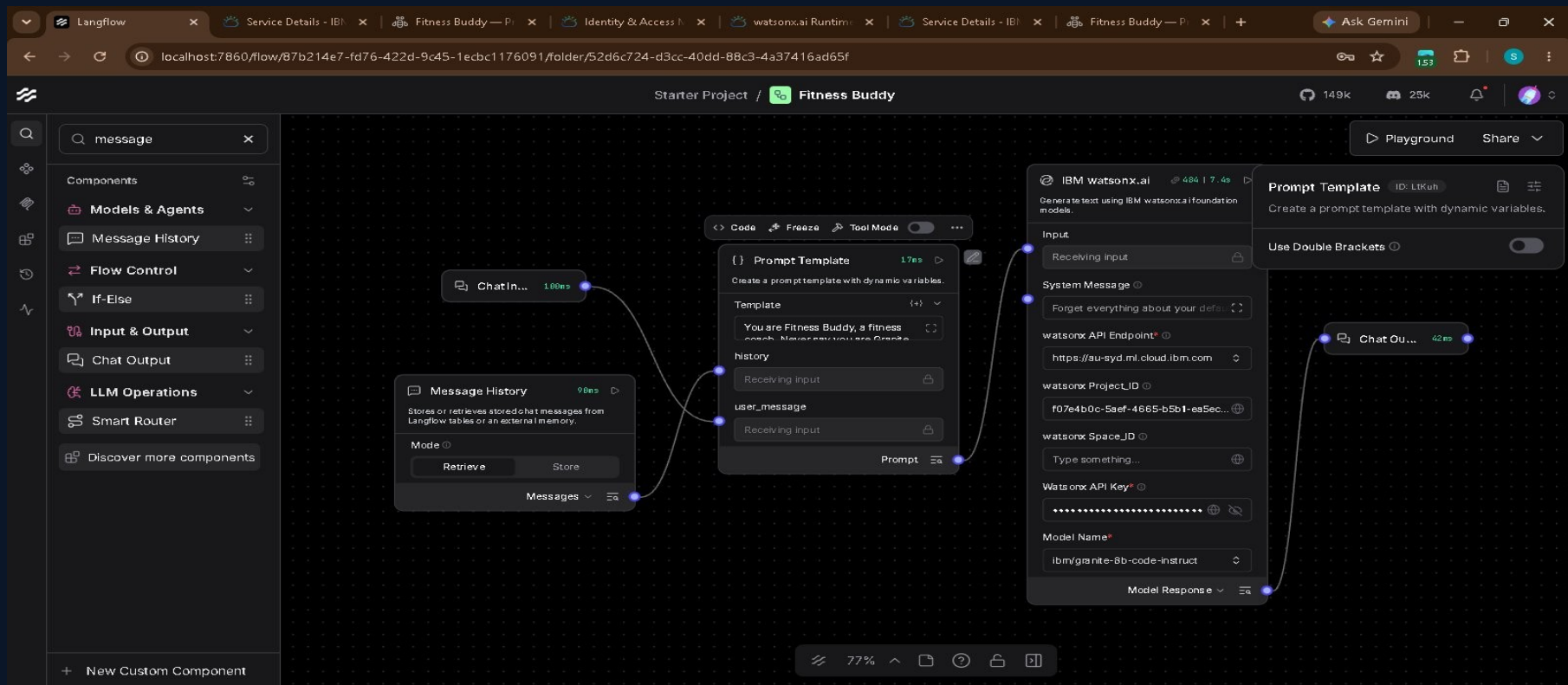
Memory component that stores conversation history so Fitness Buddy remembers user name, goals and context

5

Chat Output

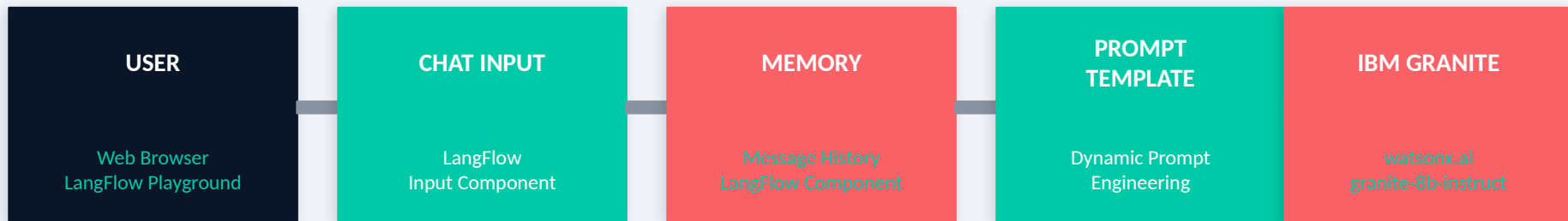
Displays AI-generated workout plans, meal suggestions, motivation tips and fitness guidance to the user

LANGFLOW WORKFLOW



LangFlow Canvas: Chat Input → Message History → Prompt Template → IBM watsonx.ai (Granite) → Chat Output

ARCHITECTURE BLUEPRINT



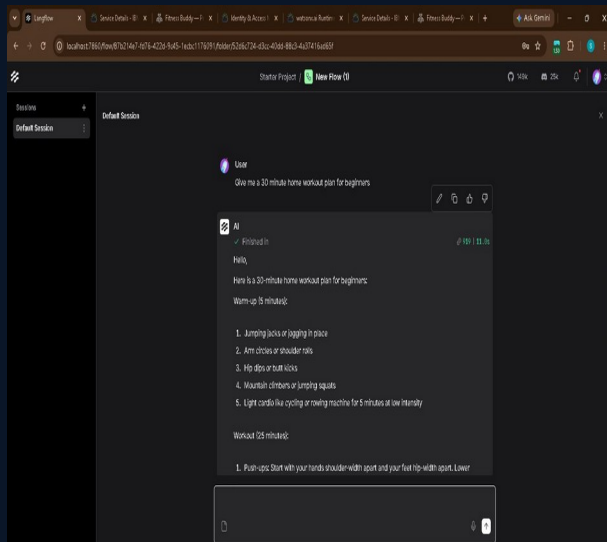
IBM CLOUD INFRASTRUCTURE

IBM Cloud Lite Account • watsonx.ai Studio • watsonx.ai Runtime (Sydney au-syd) • API Key Authentication • Project ID

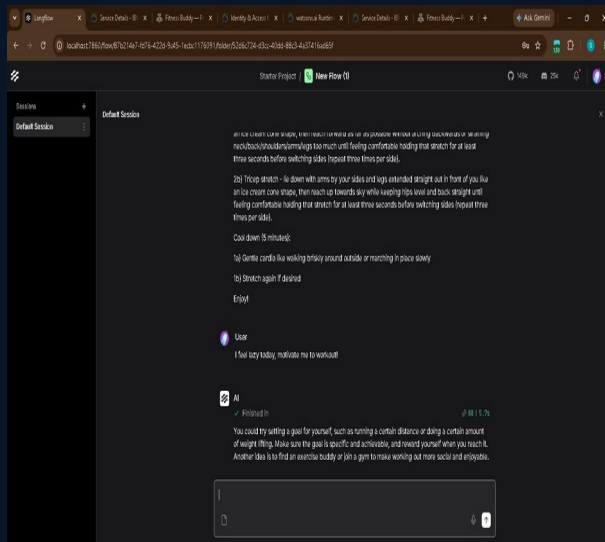


Chat Output → AI Response sent back to user → Fitness plans, meal ideas, motivation displayed

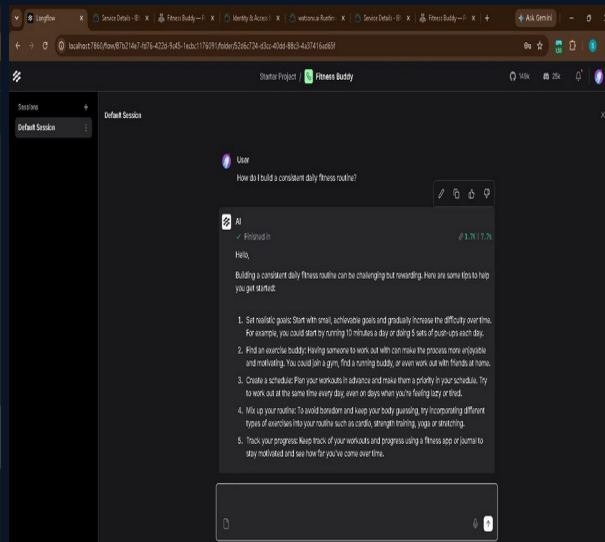
PROJECT DEMO — OUTPUT SCREENSHOTS



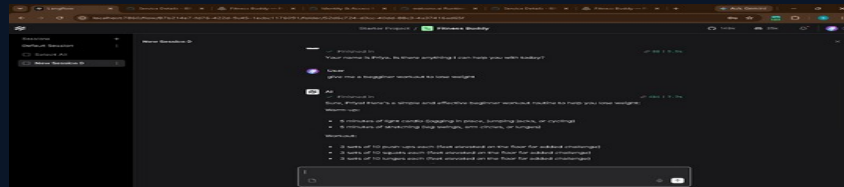
Workout Plan Output



Motivation Response



Daily Fitness Routine



ROLE OF AGENTIC AI IN FITNESS BUDDY



Intelligent Understanding

IBM Granite model understands user intent, fitness goals, and health context to generate meaningful responses rather than generic answers



Personalization Engine

Fitness Buddy adapts its recommendations based on user-provided fitness level, goals, and preferences for hyper-personalized guidance



Memory & Context Awareness

Message History component enables Fitness Buddy to remember user name, goals and conversation context across the entire session



Autonomous Decision Making

The AI autonomously decides what type of response to generate — workout plan, meal advice, or motivation — based on user query



Goal-Oriented Behavior

Acts like a proactive digital fitness coach, always driving users toward their fitness goals with structured plans and encouragement

NOVELTY & UNIQUENESS



Conversational AI Coach

Unlike static fitness apps, Fitness Buddy uses natural language conversation powered by IBM Granite for dynamic guidance



Memory-Enabled Chat

Unique conversation memory using LangFlow's Message History — remembers user context throughout the entire session



Prompt Engineering

Custom system prompts make IBM Granite behave specifically as a fitness coach — not a generic AI assistant



IBM Enterprise AI

Built on IBM watsonx.ai — enterprise-grade AI infrastructure ensuring reliability, governance and scalability



Zero-Cost Solution

Completely free using IBM Cloud Lite plan and LangFlow open-source — accessible to everyone without subscriptions



24/7 Availability

Always-on AI fitness coach available anytime — no appointments, no expensive gym memberships required

GITHUB REPOSITORY



github.com/saipriyameduri/fitness-buddy-ibm-langflow

Public Repository • MIT License • README included

Files Uploaded to Repository:



Fitness_Buddy.json

Complete LangFlow workflow export file



Screenshots/

Demo screenshots of all chatbot responses



README.md

Project description, setup instructions and flow architecture



Presentation.pptx

This project presentation for final submission

FUTURE SCOPE

01

Wearable Device Integration

Connect with smartwatches and fitness trackers to collect real-time health data (steps, heart rate, calories burned) and provide dynamic, activity-based fitness recommendations

02

Voice Assistant & Multilingual Support

Add voice interaction and support for multiple languages including Telugu, Hindi and Tamil — making Fitness Buddy accessible to diverse users across India

03

Progress Tracking Dashboard

Build a visual dashboard showing weekly workout progress, calorie tracking, fitness milestones and goal completion charts using IBM Analytics

04

RAG-Based Nutrition Database

Integrate Retrieval-Augmented Generation with verified nutrition databases (USDA, WHO) to provide evidence-based meal recommendations and dietary advice



THANK YOU!

Fitness Buddy — AI Fitness Coach

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[Click here to view GitHub Repository](#)